

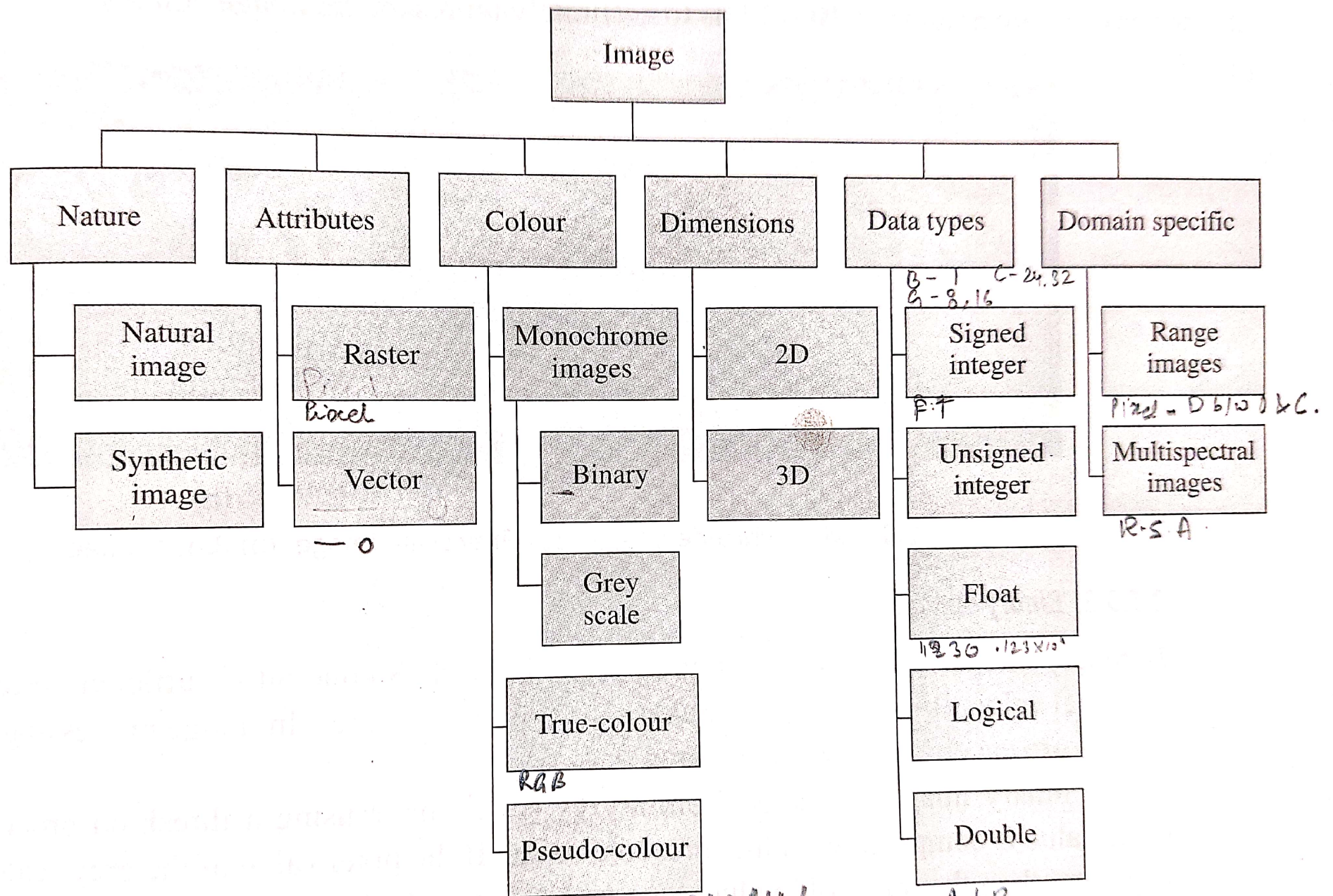


Fig. 1.4 Classification of images

Example 1.1 What is the storage requirement for a 1024×1024 binary image?

Solution For a binary image, one bit is sufficient for representing the pixel value. So the number of bits required will be $1024 \times 1024 \times 1 = 10,48,576$ bits $= 1,31,072$ bytes $= 131.072$ Kb (Assume 1 Kb $= 1000$ bytes).

Example 1.2 What is the storage requirement for a 1024×1024 24-bit colour image?

Solution Since colour images are three-band images (red, green, and blue components), the storage requirement is $1024 \times 1024 \times 3$ bytes $= 31,45,728$ bytes. If it is assumed that 1 Kb is 1000 bytes, the storage requirement is 3,145.728 Kb.

Example 1.3 A picture of physical size 2.5 inches by 2 inches is scanned at 150 dpi. How many pixels would be there in the image?

Solution The relation between the physical dimensions and the spatial resolution is simple. The pixel dimensions are obtained by multiplying the physical width and height by the scanned resolution. Therefore, the pixel dimension is as follows.

$$\begin{aligned} & (2.5 \times 150) \times (2 \times 150) \\ & = 375 \times 300 = 112500 \text{ pixels would be present} \end{aligned}$$

Example 1.4 If a 375×300 grey-scale image needs to be sent across the channel of capacity 28 kbps, then how much transmission time is required?

Solution If the picture is grey scale, then 8 bits are used. Therefore, transmission time would be

$$= \frac{375 \times 300 \times 8}{28 \times 1000} = \frac{112500 \times 8}{28000} = 32.143 \text{ sec}$$

Example 1.5 Given a grey-scale image of size 5 inches by 6 inches scanned at the rate of 300 dpi, answer the following:

- (a) How many bits are required to represent the image?
- (b) How much time is required to transmit the image if the modem is 28 kbps?
- (c) Repeat the aforementioned if it were a binary image.

Solution

- (a) Number of bits required to represent grey-scale image (uses 8 bits)

$$= 5 \times 300 \times 6 \times 300 \times 8 = 1500 \times 1800 \times 8 = 21600000 \text{ bits}$$

- (b) Total time taken to transmit image

$$= \frac{\text{Total number of bits in image}}{\text{Transmission Speed}} = \frac{21600000}{28000} = 771.43 \text{ sec}$$

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(c) If it is binary image, then the number of bits required to represent binary image

$$= 5 \times 300 \times 6 \times 300 \times 1 = 1500 \times 1800 \times 1 = 2700000 \text{ bits}$$

$$\text{The total transmission time would be} = \frac{\text{Total number of bits}}{\text{Transmission speed}} = \frac{2700000}{28000} = 96.429 \text{ sec}$$
